NOVENTRA

E-Commerce Analytics Suite | Business Insights Report

FY 2011-2014 | Prepared by: Aditya Sharma, Portfolio Project (Data Analyst) | SQL Server | Power BI | DAX

$61.8M	795	51,290	7
TOTAL REVENUE	CUSTOMERS	ORDERS	MARKETS

01 PROJECT OVERVIEW

About This Project

Noventra is a fictional global e-commerce retailer specialising in furniture, technology, and office supplies. This project demonstrates a complete end-to-end data analyst workflow: Excel cleaning, SQL Server normalisation and business analysis, and interactive Power BI dashboard delivery.

Business Problems Addressed

- **Revenue Health:** Is revenue growth genuine or driven by volume at the cost of unit value?
- **Customer Behaviour:** Who are our most valuable customers and which segments are at risk of churning?
- **Product Performance:** Which products drive revenue and where is discounting eroding margin?
- **Operational Efficiency:** Where are shipping delays concentrated and what is causing them?
- **Market Opportunity:** Which geographic markets are growing fastest and where should we invest?

Technology Stack

LAYER	TOOL	PURPOSE
Data Cleaning	Microsoft Excel	Remove duplicates, standardise formats, fix discount % field, handle nulls
Storage & Analysis	SQL Server (SSMS)	3NF normalisation, analytical views, stored procedures, window functions, triggers
Visualisation	Power BI Desktop	DAX measures, star schema relationships, interactive dashboards, field parameters

02 SQL SERVER DATABASE DESIGN & ANALYSIS

2.1 Data Architecture: Star Schema (3NF)

The raw flat-file dataset was normalised into a star schema with one fact table and four dimension tables to eliminate data redundancy, enforce referential integrity, and enable efficient analytical queries.

TABLE	TYPE	PRIMARY KEY	PURPOSE
orders	Fact	order_line_id (surrogate)	Revenue, quantity, discount, delivery days
customers	Dimension	customer_id	Customer name, segment, age
products	Dimension	product_id	Product name, sub-category, category
locations	Dimension	location_id	State, country, market, region
dates	Dimension	date_value	Year, quarter, month, day time intelligence

Design Note

order_id repeats across multiple product line items within one order, so it cannot serve as a primary key. A surrogate key (order_line_id IDENTITY) was added to uniquely identify each line item - standard fact table design in data warehousing.

2.2 Analytical Views

VIEW	BUSINESS QUESTION	KEY TECHNIQUE	POWER BI
vw_product_performance	Top 25 products by revenue?	DENSE_RANK(), GROUP BY	Yes
vw_customer_segment_insights	Which segment drives most revenue?	CASE WHEN age banding	Yes
vw_shipping_logistics	Which ship mode is most efficient?	DENSE_RANK(), AVG days	Future
vw_monthly_revenue	What is MoM revenue growth?	LAG() window function	Ref
vw_yearly_revenue	What is YoY annual growth?	LAG() window function	Ref
vw_customer_lifetime_value	Top lifetime value customers?	DENSE_RANK() on spend	Yes
vw_repeat_purchase_analysis	Repeat purchase rate by segment?	CASE WHEN order count > 1	Yes
vw_rfm_segmentation	Champions vs At Risk customers?	NTILE(4) on Recency, Freq, Mon	Yes

2.3 RFM Customer Segmentation

RFM scoring uses NTILE(4) quartile ranking—fully data-driven. Each customer receives a score of 1-4 on Recency, Frequency, and Monetary independently. Combined scores determine the segment label.

DIMENSION	SCORE 4 (BEST)	SCORE 1 (WORST)
Recency (R)	Most recently active	Longest since last order
Frequency (F)	Highest number of orders	Fewest orders placed
Monetary (M)	Highest total lifetime spend	Lowest total lifetime spend

2.4 Stored Procedures

PROCEDURE	PARAMETER	BUSINESS USE
sp_customer_history	@customer_id INT	Full order history per customer account management and dispute resolution
sp_sales_between_dates	@start_date, @end_date	Revenue and order count for any custom date range - period comparison reporting
sp_customer_churn_analysis	@days INT	Customers inactive for N+ days. Uses fixed date 2014-12-31 (not GETDATE) to match dataset end.

2.5 Triggers and Audit Trail

Three AFTER triggers (INSERT, UPDATE, DELETE) on the orders table write every modification to an order_audit table with timestamp - demonstrating data governance, change tracking, and enterprise data management practices.

03 KEY BUSINESS FINDINGS

All figures derived directly from the Noventra dataset. No values estimated or fabricated.

3.1 Executive Revenue Summary

METRIC	2013	2014	YOY	SIGNAL
Total Revenue	$16,743,061	$20,985,128	+25.3%	**Positive**
Total Orders	6,721	8,531	+26.9%	**Positive**
Avg Order Value (AOV)	$2,491.16	$2,459.87	-1.3%	**Risk**
Avg Selling Price (ASP)	$347.83	$346.16	-0.5%	**Risk**
Units Sold	48,000	61,000	+25.9%	**Positive**
Avg Discount	14.03%	14.33%	+25.9%	**Risk**
Discount Revenue Loss	$1.75M	$2.21M	+26.5%	**Risk**

Core Finding

Revenue grew +25.3% and orders grew +26.9% - but AOV fell -1.3% and ASP fell -0.5% simultaneously. Growth is entirely volume-driven. With $2.21M sacrificed to discounts (up 26.5% YoY), the business may be buying order volume through discounting rather than generating genuine demand. This is a **growth quality risk**.

3.2 Customer Intelligence

METRIC	VALUE	IMPLICATION
Total Customers	795	All repeat buyers; 100% repeat rate across 4 years
Consumer Revenue Share	51.2% ($31.7M)	Largest segment by total revenue volume
Corporate CLV/Customer	$79,034	Highest value per customer; above Consumer at $77,389
Top Customer	Tom Ashbrook	$226,424 lifetime revenue
Highest Revenue Age Group	Gen X (41-60)	38.5% of total revenue ($23.8M)
Lowest Revenue Age Group	Gen Z (under 25)	Only 9.9% of revenue - $6.1M
Peak Revenue Month	November 2014	$2,686,570 - Q4 seasonality is significant

3.3 Product Performance and Margin

METRIC	VALUE	IMPLICATION
Technology Revenue	37.3% ($23M)	Leading category, closing gap vs Furniture (32.5%) and Office Supplies (30.2%)
Highest Rev/Unit	Tables ($1,186)	Premium margin product - under-leveraged in marketing and inventory focus
Lowest Rev/Unit	Labels ($41)	High volume, negligible unit value; bundling or delisting candidate
Top Product	Apple Smart Phone	$432,251; 4 of top 5 revenue products are smartphones
Orders > 20% Discount	25.2%	One in four orders is heavily discounted; significant margin exposure
Furniture Disc Rate	46.66%	Highest high-discount concentration of any category

3.4 Operations and Logistics

METRIC	VALUE	IMPLICATION
Overall Delay Rate	18.2%	9,312 orders delayed beyond 5 days annually
On-Time Rate	81.8%	4 in 5 orders delivered within 5 days
Standard Class Delay	**30.3%**	Only shipping method generating delays; all other methods at 0%
First/Second/Same Day	0.0%	Perfect on-time delivery across all three methods
Low Priority Avg Days	6.5 days	Exceeds 5-day threshold; primary delay driver by priority classification
Critical Priority Days	1.8 days	Fastest fulfilment tier; strong SLA performance

Sharpest Finding

Standard Class is the ONLY shipping method generating delays (30.3% delay rate). First Class, Second Class, and Same Day all have exactly 0% delay rates. The entire 18.2% delay problem is isolated to a single shipping method — a specific, actionable carrier management finding.

3.5 Regional Market Analysis

MARKET	REVENUE	SHARE	YOY GROWTH	AOV
APAC	$18.3M	29.5%	+26.7%	$3,358
EU	$14.8M	23.9%	+34.9%	$3,221
US	$11.5M	18.6%	+17.8%	$2,293
LATAM	$10.6M	17.1%	+17.4%	$2,062
EMEA	$3.3M	5.4%	+29.7%	$1,347
Africa	$3.1M	5.1%	+24.3%	$1,402
Canada	$237K	0.4%	+16.0%	$1,181

04 POWER BI DASHBOARD PAGE GUIDE

Page 1: Executive Revenue & Strategic Planning

Business question: Is the business growing, and is the growth healthy?

VISUAL	TYPE	BUSINESS PURPOSE
Revenue/Orders/AOV/ASP Cards	Card x4	Green/red arrows signal health instantly. AOV and ASP in red flag the growth quality risk on landing.
ASP% Over Months	Line chart	Tracks unit price monthly; confirms whether ASP decline is seasonal or structural.
Revenue & YoY % Over Months	Combo chart	Revenue bars vs YoY growth line shows whether growth is accelerating or decelerating month by month.
Orders Based on Priority	Donut chart	Priority mix - 57.4% Medium priority contextualises the Operations shipping delay analysis.
Revenue Breakdown by Category	Decomp tree	Drill from Category to Sub-Category to Product identifies exact revenue drivers at any level.
Adjusted Price % Simulator	What-If param	Scenario planning; models revenue impact of price changes without touching real data.

Page 2: Customer Intelligence

Business question: Who are our customers, how do they behave, and who is at risk?

VISUAL	TYPE	BUSINESS PURPOSE
Active Customers / Repeat Rate / CLV / At Risk	Card x4	Customer health snapshot; repeat rate and CLV confirm loyalty and segment value.
RFM Segment Distribution	Horizontal bar	Classifies 795 customers into Champions, Loyal, At Risk, Hibernating, New, Regular using NTILE(4).
Repeat vs Single Buyers Table	Matrix	~100% repeat rate per segment confirms consistent cross-segment loyalty pattern.
Revenue by Customer Segment	Donut chart	Consumer 51.2%, Corporate 30.4%, Home Office 18.4% drives segment investment prioritisation.
Metric by Age Group	Bar + parameter	Dynamic switching between Revenue, Orders, Customers, AOV. Gen X = 38.5% of revenue.
AOV Trend Over Years With Customers	Combo chart	Customer base grew 75% from 2011-2014 while AOV held stable; volume growth without value dilution.

Page 3: Product Intelligence & Margin Analysis

Business question: What drives revenue and where is discounting eroding margin?

VISUAL	TYPE	BUSINESS PURPOSE
Top Category / Discount Revenue Loss / Units / Avg Disc	Card x4	$2.21M revenue lost to discounts in 2014 is the headline margin finding - visible on landing.
Avg Discount % and Revenue Over Sub-Category	Scatter chart	Bubbles = sub-categories. Reference line at 14.2% creates 4 quadrants: Premium, Discount Dependent, Niche, Margin Risk.
Order Discount Based on Category	Stacked bar	Discount distribution per category. Furniture carries 46.66% high-discount orders — highest of any category.
CY Revenue by Month and Category	Line chart	Technology peaks sharply in Q4. Furniture and Office Supplies show steadier year-round demand.
Top N Products With Revenue	Treemap + slicer	Dynamic top 3-25 product view. 4 of top 5 products are smartphones — concentration risk visible immediately.

05 STRATEGIC RECOMMENDATIONS

Revenue & Pricing

- **Investigate whether discounting drives order volume growth:** 25.2% of orders carry discounts above 20%.
- **Introduce bundling and upsell strategies** to lift AOV without relying on additional order volume.
- **Build Q4 inventory and campaign strategy** around November peak ($2.69M) - Technology drives the sharpest spike.
- **Monitor discount revenue loss trend:** $2.21M in 2014 up 26.5% from $1.75M in 2013; without AOV growth this signals discount dependency.

Customer Strategy

- **All 795 customers are repeat buyers:** Shift spend from acquisition to increasing revenue per customer.
- **Corporate CLV per customer ($79,034) is highest:** Evaluate dedicated account management or loyalty programme.
- **Use RFM segmentation for targeted campaigns:** Reward Champions, re-engage At Risk and Hibernating customers immediately.
- **Gen Z (under 25) represents only 9.9% of revenue:** Assess whether this is a product-market fit gap or an untapped opportunity.

Product & Margin

- **4 of top 5 revenue products are smartphones:** Diversify Technology sub-category to reduce supplier concentration risk.
- **Prioritise Copiers ($963/unit) and Tables ($1,186/unit)** in marketing for highest margin contribution per unit.
- **Conduct discount ROI analysis on Furniture:** 46.66% high-discount order rate must be validated against incremental revenue.

- **Evaluate Labels ($41/unit) and Fasteners ($49/unit)** for bundling or delisting - high volume with negligible unit value.

Operations & Logistics

- **Standard Class is the sole source of all delays (30.3%):** Audit carrier SLAs and escalate performance failures; review priority classification at order entry. First, Second, and Same Day methods all have 0% delay rates.
- **Low priority orders average 6.5 days, exceeding the 5-day threshold:** Route time-sensitive Low priority orders to Second Class or First Class.
- **Quantify repeat-purchase impact** of 9,312 annual delayed orders to build the carrier investment business case.

Market Investment

- **EU grew +34.9% YoY with AOV of $3,221:** Fastest growing premium market; increase investment immediately.
- **APAC has highest AOV ($3,358) and largest revenue share (29.5%):** Protect through service quality and retention.
- **Canada generates only $237K (0.4% of revenue) after 4 years:** Conduct a formal market viability review.
- **LATAM growing at +17.4% vs EU at +34.9%:** Assess whether gap reflects market maturity or a competitive issue.